Review via Linear Regression

02-680: Essentials of Mathematics and Statistics

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1 Linear Regression

The ***linear regression*** problem is as follows (we covered this briefly in a previous topic (Independence), but with a slightly different formula):

x — Independent variable[s] (what we control)

y — Dependent variable[s] (the measured values)

β_s – slope

β_0 – intercept

Some examples of regression in biology:

Independent	Dependent
Expression of genes expressed in immune response	asthma severity
Expression of regulator genes	Expression of target genes
Clinical variables	Insulin level
...	...

Lets assume we have some $x, y \in \mathbb{R}^n$, for n samples where each (x_i, y_i) are corresponding. Then

$$y = x^T \beta_s + \beta_0$$

To simplify things lets define: $X = \begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix}$ and $\beta = \begin{bmatrix} \beta_s \\ \beta_0 \end{bmatrix}$, then the equation above becomes simply:

$$y = X\beta.$$

Recall (possibly) from earlier that we want to find β that minimizes the $\|y - X\beta\|_2^2$, or the square of the L_2-norm.

$$\begin{aligned}\|y - X\beta\|_2^2 &= (y - X\beta)^T(y - X\beta) \\ &= (y^T - \beta^T X^T)(y - X\beta) \\ &= y^T y + \beta^T X^T X\beta - 2\beta^T X^T y\end{aligned}$$

To find the critical point of the curve (i.e. the minimum) we need

$$\begin{aligned}\frac{d}{d\beta} y^T y + \beta^T X^T X\beta - 2\beta^T X^T y &= 0 \\ \frac{d}{d\beta}\beta^T X^T X\beta - \frac{d}{d\beta} 2\beta^T X^T y &= 0 \\ 2X^T X\beta - 2X^T y &= 0 \\ 2X^T X\beta &= 2X^T y \\ \beta &= (X^T X)^{-1} X^T y\end{aligned}$$

When does this solution exist?

The last step needs $X^T X$ to be invertible, and eigenelements (Topic 11) tell us when it is.

- $X^T X$ is symmetric, so its eigenvalues are real.
- A matrix is singular exactly when its determinant is 0, and the determinant is the product of the eigenvalues. So $X^T X$ is invertible if and only if ***none*** of its eigenvalues is 0.
- A zero eigenvalue happens when the columns of X are linearly dependent (Topic 10). For example, if every x_i is the same value then the first column of X is a multiple of the column of 1s. With several predictors, the same thing happens if one is a combination of the others (e.g., one measurement is the sum of two others). Then $X^T X\beta = X^T y$ has many solutions and $(X^T X)^{-1}$ does not exist.
- For any eigenvector v of $X^T X$ with eigenvalue λ, we have $\lambda\|v\|_2^2 = v^T X^T X v = \|Xv\|_2^2 \geq 0$, so every eigenvalue is non-negative. This is why the critical point we found is a minimum of the squared error and not a maximum.

What if I want to know the probability of some y_i given x_i? (so $p(y_i \mid x_i)$)

Assuming its some natural process, we can assume

$$p(y_i \mid x_i) \sim \mathcal{N}(\beta_0 + \beta_s x_i, \sigma^2).$$

or equivalently

$$y = \beta_0 + \beta_s x_i + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2).$$

So given the data, we can actually use MLE to find β!

$$\begin{aligned}
\ln p(\mathcal{D} \mid \beta, \sigma) &= \ln \left[\prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma}(y_i - x_i\beta)^2} \right] \\
&= \sum_{i=1}^{n} \ln \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma}(y_i - x_i\beta)^2} \right] \\
&= \frac{-n}{2} \ln \sigma^2 + \sum_{i=1}^{n} \ln e^{-\frac{1}{2\sigma^2}(y_i - x_i\beta)^2} \\
&= \frac{-n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - x_i\beta)^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial\beta} \frac{-n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - x_i\beta)^2 &= \frac{\partial}{\partial\beta} - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - x_i\beta)^2 \\
&= \frac{\partial}{\partial\beta} - \frac{1}{2\sigma^2} (Y - X\beta)^2 \\
&= -\frac{1}{2\sigma^2} \frac{\partial}{\partial\beta} (Y^2 - 2Y^T X\beta + \beta^T X^T X\beta) \\
&= ...
\end{aligned}$$

$\hat{\beta}_{MLE} = (X^T X)^{-1} X^T y$, that's just Least Squares!

But we don't know σ, that's actually, what we want to know because it models how uncertain we are with any given output.

$$\frac{\partial}{\partial\sigma^2} \frac{-n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - x_i\beta)^2 = \frac{-n}{2\sigma^2} + \frac{1}{2} (\sigma^2)^{-2} \sum_{i=1}^{n} (y_i - x_i\beta)^2$$

$\hat{\sigma}^2_{MLE} = \frac{\sum_{i=1}^{n} (y_i - x_i\beta)^2}{n}$

With these we can determine for some new $\tilde{y}$ and $\tilde{x}$ what $p(\tilde{y} \mid \tilde{x})$ is.